

Data Types, Variables, Operators

Study Guide

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2.1.1 Java Data Types and Variables

In Java, **data types** specify the type of data a variable can hold. Java is a **strongly typed language**, meaning every variable must be declared with a data type before it is used.

Java data types are classified into **two main categories**:

◆ 1. Primitive Data Types

These are the **basic building blocks** of data manipulation. Java has **8 primitive types**:

Data Type	Description	Size	Default Value	Example
byte	8-bit signed integer	1 byte	0	byte b = 10;
short	16-bit signed integer	2 bytes	0	short s = 100;
int	32-bit signed integer	4 bytes	0	int x = 500;
long	64-bit signed integer	8 bytes	0L	long l = 100L;
float	32-bit floating point	4 bytes	0.0f	float f = 10.5f;
double	64-bit floating point	8 bytes	0.0d	double d = 99.99;
char	Single 16-bit Unicode	2 bytes	'\u0000'	char c = 'A';
boolean	Stores true or false	~1 bit	false	boolean flag = true;

◆ 2. Non-Primitive (Reference) Data Types

These are **user-defined** or built-in types that refer to objects and support more complex operations.

Type	Description	Example
String	A sequence of characters	String name = "Java";
Array	Collection of similar type data elements	int[] arr = {1,2,3};
Class	User-defined blueprint for objects	class Student { ... }
Interface	Abstract type with abstract methods	interface Shape { ... }
Object	Base class of all Java classes	Object obj = new Object();

2.1.2 Variables in Java

In Java, a **variable** is a **named memory location** used to store data that can be changed during the execution of a program. Every variable in Java must be declared with a **data type** before it is used.

◆ Variable Declaration and Initialization

✎ Syntax:

```
dataType variableName;           // Declaration
variableName = value;             // Initialization
dataType variableName = value;    // Both
```

✓ Example:

```
int age = 25;
String name = "Alice";
boolean isActive = true;
```

◆ Variable Naming Rules in Java

- Must start with a **letter, \$, or _** (not a digit).
- Cannot use Java **reserved keywords** (class, int, etc.).
- **Case-sensitive** (Age and age are different).
- Use meaningful names (int score is better than int x).

◆ 3. Dynamic Initialization of Variables

Assigning values to variables **at runtime**, often using **expressions** or **input**.

✓ Examples:

```
int a = 10, b = 20;
int sum = a + b;    // Initialized using expression
Scanner sc = new Scanner(System.in);
int age = sc.nextInt(); // Initialized using user input
```

2.2.1. Scope and Lifetime of Variables

◆ Types of Variables in Java

Java variables are classified into **three main types** based on their placement and scope:

1. Local Variables

- Declared inside a method, constructor, or block.
- **Only accessible** within that method or block.
- Must be **initialized before use**.
- Example:

```
void greet() {  
    String message = "Hello!";  
    System.out.println(message);  
}
```

2. Instance Variables

- Declared **inside a class** but **outside any method**.
- Each object of the class has its **own copy**.
- Also called **non-static fields**.
- Have **default values** if not initialized.
- Example:

```
public class Student {  
    int rollNo;    // Instance variable  
    String name;   // Instance variable  
}
```

3. Static Variables (Class Variables)

- Declared with the static keyword inside a class.
- Shared **among all objects** of the class.
- Initialized only once at class loading time.
- Example:

```
public class Counter {  
    static int count = 0; // Static variable  
}
```

◆ Variable Visibility and Block-Level Scope

💡 Scope Definition:

Scope refers to **where in the program a variable is accessible**.

Types of Scopes:

Scope	Where it's declared	Visible to
Block Scope	Inside {} blocks	Inside that block only
Method Scope	Inside a method	Only within that method
Class Scope	At class level (instance/static)	All class methods (based on access)

◆ Lifetime of Variables

Variable Type	Lifetime	Memory Area
Local	From entry to exit of method/block	Stack
Instance	From object creation to object destruction	Heap
Static	From class loading to program end	Method Area

 Practice Activity :

- Identify each variable type.
- Predict the lifetime and scope.

```
public class Demo {  
    static int counter = 0;  
  
    int id;  
  
    public Demo() {  
        id = ++counter;  
    }  
  
    public void print() {  
        String localMessage = "Hello from object " + id;  
        System.out.println(localMessage);  
    }  
}
```

➤ Summary of Key Concepts:

◆ Java supports two categories of data types:

1. Primitive Data Types (8 total)
 - Integer types: byte, short, int, long
 - Floating types: float, double
 - Others: char (character), boolean (true/false)
2. Non-Primitive (Reference) Data Types
 - Examples: String, Array, Class, Interface, Object
 - Store references to objects, not actual data values.

◆ A variable is a container that holds a value of a specific data type.

Types of variables:

- **Local** – Declared inside methods or blocks; exists temporarily.
- **Instance** – Declared in a class but outside methods; belongs to each object.
- **Static** – Declared with static; shared among all objects of a class.

➤ References:

❑ Official Java Documentation (Oracle):

- [Java Variables – Oracle](#)
- [Java Data Types and Variables](#)

❑ Books:

- *“Java: The Complete Reference”* by Herbert Schildt (11th Edition)
- *“Head First Java”* by Kathy Sierra and Bert Bates

❑ Online Java Learning Platforms:

- GeeksforGeeks – Variable Scope in Java
- JavaTPoint – Java Variable Types
- W3Schools – Java Variables

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